

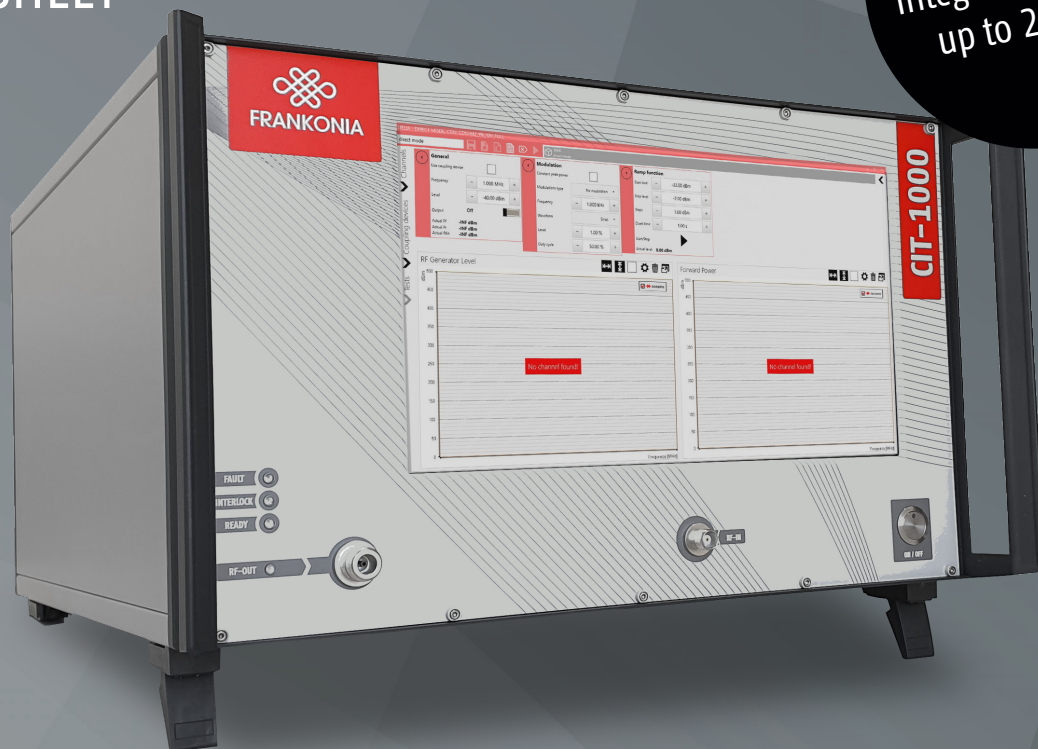


FRANKONIA
EMC Test-Systems GmbH

CIT-1000

DATA SHEET

NEW:
Integrated amplifier
up to 200 Watt



CONDUCTED IMMUNITY TEST SYSTEM
FOR IEC 61000-4-6, ISO 11452-4 AND MIL-STD-461 CS114

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DESCRIPTION

The CIT-1000 is a further development of the for years available and proven „CIT-10“, with the extended frequency range up to 400 MHz (signal generator and power meter up to 1200 MHz), stand-alone operation via integrated touchscreen PC as well as integrated directional coupler and power meter for forward and reverse power measurement.

As usual with our CIT-series, all integrated instruments, like signal generator, RF power amplifier and the 3-channel RF power meter can be used as „stand-alone-unit“, too.

Hence, the signal generator and the RF power meter can also be used for radiated immunity tests acc. IEC/EN 61000-4-3.

Furthermore an additional external RF power amplifier could be connected to the CIT-1000 for this purpose.

SPECIAL FEATURES

- ✓ Conducted RF immunity tests acc. to IEC/EN 61000-4-6, BCI-tests acc. to ISO 11452-4 and MIL-STD 461 (CS 114)
- ✓ Signal generator, RF power amplifier, 3-channel power meter and directional coupler combined in one 19“-case.
- ✓ Stand-alone operation via integrated touchscreen PC
- ✓ Frequency range: 4 kHz – 1.2 GHz
- ✓ With integrated amplifier 25W / 75W / 200W
- ✓ Control software included
- ✓ Temperature measuring input for control and display of the BCI clamp temperature
- ✓ Interfaces: USB, LAN, GPIB (optional)

TECHN. SPECIFICATIONS

RF power amplifier (version)	25 W	75 W MIL	75 W	200 W
Frequency range	100 kHz – 250 MHz	10 kHz – 400 MHz	100 kHz – 400 MHz	100 kHz – 400 MHz
Output power nominal	25 W	75 W	75 W	200 W
Linear @ 1 dB compression	20 W	50 W	50 W	100 W
Gain	46 dB nominal	51 dB nominal	51 dB nominal	51 dB nominal
Flatness	±1.5 dB maximum			
Input power for rated output	1 mW / 0 dBm			
Input / output impedance	50 Ω			
Input VSWR	1.5:1 max			
Harmonic distortion	< -20 dBc @ 20 W	< -20 dBc @ 50 W	< -20dBc @ 50 W	< -20 dBc @ 100 W
Noise figure	typ. 5 dB	typ. 7 dB	typ. 7 dB	tbd
Spurious output	<-75 dBc @ 10 W			



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TECHNICAL SPECIFICATIONS

RF-Generator

Two switchable outputs (Only one can be used simultaneously)	2 x SMA
Frequency range	9 kHz to 1.2 GHz (usable from 4 kHz)
Frequency resolution	1 Hz
Output level range	0 to -63 dBm
Output level resolution	0.1 dB
Harmonics	< 30 dBc
Spurious	< 45 dBc
Amplitude mod. (internal)	0 - 100%, resolution 1%
Amplitude mod. (external)	0-100%, max. Amplitude 1V=100%, BNC jack

Pulse modulation (internal)	5-95%, resolution 1%
Pulse modulation (external)	DC...1MHz, 3,3/5V CMOS/TTL, BNC jack

LF-Generator (modulation)

Connector	BNC jack
Frequency range	1 Hz to 100 kHz
Frequency resolution	0.1 Hz
Signal	Sine wave / square wave / triangular
Amplitude	0...1 V

RF-Voltmeter 1 (test-level)

Connector	BNC jack
Frequency range	9 kHz to 1.2 GHz (usable from 4 kHz)
Measuring range	-40 to +30 dBm

TECHNICAL SPECIFICATIONS

RF-voltmeter 2+3 (fwd. and rev. power)

Connector	2 x SMA
Frequency range	9 kHz to 1.2 GHz (usable from 4 kHz)
Measuring range	-40 to + 33 dBm + direct. coupler typ. 40 dB

EUT-Monitor input

Input voltage	0 to 10 V DC
Resolution	2.5 mV
Input impedance	100 k Ω

EUT-failed input

Input signal	0 to 10 V DC
Detection mode	2.5 mV
Temperature measurement	10 to 100 °C (1039 to 1385 Ω) resolution < 1 °C (PT 1000)

SCPI interfaces

USB 2.0	USB-B
LAN, 100 Mbit	RJ45
GPIB (optional)	Centronics

Digital I/Os

Out	4 Bit Digital out, 5 V CMOS/TTL
In	4 Bit Digital in, 5 V CMOS/TTL

Interlock

Closes at	R < 1 Ω
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